

INTERNATIONAL STANDARD

**Optical fibre cables –
Part 3-40: Outdoor cables – Family specification for sewer cables and conduits
for installation by blowing and/or pulling in non-man accessible storm and
sanitary sewers**



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INTERNATIONAL ELECTROTECHNICAL COMMISSION

OPTICAL FIBRE CABLES –

Part 3-40: Outdoor cables – Family specification for sewer cables and conduits for installation by blowing and/or pulling in non-man accessible storm and sanitary sewers

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International Standard IEC 60794-3-40 Ed. 1 has been prepared by sub-committee 86A: Fibres and cables, of IEC technical committee 86: Fibre optics.

This standard is to be used in conjunction with IEC 60794-1-1, IEC 60794-1-2 and IEC 60794-3.

The text of this standard is based on the following documents:

FDIS	Report on voting
86A/1228/FDIS	86A/1241/RVD

Full information on the voting for the approval of this standard can be found in the report on voting indicated in the above table.

This publication has been drafted in accordance with the ISO/IEC Directives, Part 2.

A list of all parts of IEC 60794 series, published under the general title *Optical fibre cables*, can be found on the IEC website.

The committee has decided that the contents of this publication will remain unchanged until the maintenance result date indicated on the IEC web site under "<http://webstore.iec.ch>" in the data related to the specific publication. At this date, the publication will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

OPTICAL FIBRE CABLES –

Part 3-40: Outdoor cables – Family specification for sewer cables and conduits for installation by blowing and/or pulling in non-man accessible storm and sanitary sewers

1 Scope

This part of IEC 60794 is a family specification that covers sewer cables and conduits for installation by blowing and/ or pulling in non-man accessible storm and sanitary sewers, also applicable for man-accessible and lateral ones. Systems built with components covered by this standard are subject to the requirements of sectional specification IEC 60794-3.

Sewer cable and conduit constructions have to meet the different requirements of the sewer operating companies and/or associations regarding chemical, environmental, operational, cleaning and in general maintenance conditions.

Preferential applications, describing sewer cable characteristics versus methods of installation is reported in Annex A and Annex B for non-man accessible sewers.

Clause 4 describes a blank detail specification for sewer cables and conduits for installation by blowing and/or pulling in storm and sanitary sewers. It incorporates some minimum requirements.

Detail specifications may be prepared on the basis of this family specification.

The parameters specified in this standard may be affected by measurement uncertainty arising either from measurement errors or calibration errors due to lack of suitable standards. Acceptance criteria should be interpreted with respect to this consideration.

The number of fibres tested is representative of the sewer cable and should be agreed between the customer and the supplier.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60304:1982, *Standard colours for insulation for low-frequency cables and wires*

IEC 60793-1-20, *Optical fibres – Part 1-20: Measurement methods and test procedures – Fibre geometry*

IEC 60793-1-40, *Optical fibres – Part 1-40: Measurement methods and test procedures – Attenuation*

IEC 60793-1-44, *Optical fibres – Part 1-44: Measurement methods and test procedures – Cut-off wavelength*

IEC 60793-2, *Optical fibres – Part 2: Product specifications– General*

IEC 60793-2-50, *Optical fibres – Part 2-50: Product specifications – Sectional specification for class B single-mode fibres*

IEC 60794-1-1, *Optical fibre cables – Part 1-1: Generic specification – General*

IEC 60794-1-2, *Optical fibre cables – Part 1-2: Generic specification – Basic optical cable test procedures*

IEC 60794-3, *Optical fibre cables – Part 3: Sectional specification – Outdoor cables*

IEC 60794-3-10, *Optical fibre cables – Part 3-10: Outdoor cables – Family specification for duct and directly buried optical telecommunication cables*

IEC 60811-1-1:1993, *Common test methods for insulating and sheathing materials of electric cables and optical cables – Part 1-1: Methods for general application – Measurement of thickness and overall dimensions – Tests for determining the mechanical properties*

IEC 60811-5-1:1990, *Insulating and sheathing materials of electric and optical cables – Common test methods – Part 5-1: Methods specific to filling compounds – Drop point – Separation of oil – Lower temperature brittleness – Total acid number – Absence of corrosive components – Permittivity at 23 °C – DC resistivity at 23 °C and 100 °C*

3 Symbols

For the purposes of this document, the following symbols apply.

APL Aluminium/polyethylene laminate

SPL Steel/polyethylene laminate

λ_{CC} cabled fibre cut-off wavelength

d nominal outer diameter of the sewer cable

d_c nominal outer diameter of the conduit

DS detail specification

$n \times d$ a value times cable outer diameter used for bends, mandrels, etc.

T_O threshold tensile load below which no attenuation and/or fibre strain increase should occur in the tensile performance test

T_M the acceptable amount of short-term tensile load that can be applied to the cable without permanent degradation of the characteristics of the fibres in the tensile performance test

T_{A1} temperature cycling test low-temperature limit according to IEC 60794-1-2, method FI

T_{A2} temperature cycling test low-temperature limit according to IEC 60794-1-2, method FI

T_{B1} temperature cycling test high-temperature limit according to IEC 60794-1-2, method FI

T_{B2} temperature cycling test high-temperature limit according to IEC 60794-1-2, method FI

t_1 temperature cycling dwell time

4 Family specification for sewer cables and conduits for installation by blowing and/or pulling in sewers (blank detail specification and minimum requirements)

4.1 Construction

4.1.1 General

In addition to the constructional requirements of sectional specification IEC 60794-3, the following considerations apply to the sewer cables and/or conduits.

The sewer cables and/or conduits shall be designed and manufactured for an expected operating lifetime of at least 15 years. It shall be possible to install or remove the cable in or from the sewer throughout the operational lifetime. Upon removal of the cable or conduit, no remnants which would snag normal sewage debris or which would affect laminar sewage flow shall remain. The materials in the sewer cable and/or as well as accessories including fixing elements and conduits shall not present a health hazard within its intended use.

4.1.2 Conduits

In case of use, the conduits with outer nominal diameters ranging from xx mm to yy mm shall be able to resist pressure differences needed for installation by blowing. They shall be circular and, if needed, corrugated in cross-section throughout their length. The material shall withstand all possible chemical attacks by the sewer liquid itself, as for instance the stainless steel type 1.4401 (X5CrNiMo17-12-2), 1.4571 (X6CrNiMoTi17-12-2), AISI 316L). Inner- and outer-diameter and overall minimum wall thickness shall be specified.

4.1.3 Sewer cables

A sewer cable in accordance to this specification should be suitable for installation in non-man accessible storm and sanitary sewers by the following installation methods, also applicable for man-accessible sewers and for lateral sewer systems:

- blowing and/or pulling into a conduit, previously fixed to the sewer wall or, for laterals, placed in between the lateral sewer wall and the in-liner system;
- direct installation into the sewer duct, according to the following applications:
 - drilling and screwing to the inner wall;
 - spanning between manholes, etc. similarly to aerial cables;
 - laying on the ground of the sewer.

The attenuation of the installed cable at the operational wavelength(s) shall not exceed values agreed between the customer and the supplier.

There shall be no fibre splice in a delivery length unless otherwise agreed by the customer and the supplier.

It shall be possible to identify each individual fibre throughout the length of the sewer cable.

4.1.4 Rodent protection

Rodent protection is generally needed but there is presently no standard test to measure the level of rodent protection. Rodent protection can be provided by the cable or by the duct.

Tests and requirements are under consideration.

4.2 Optical fibres

4.2.1 Single-mode dispersion unshifted (B1.1) optical fibre

Table 1 – Single-mode dispersion unshifted (B1.1) optical fibre

Characteristics (9)	IEC 60794-3 Clause/subclause (10)	Family requirements (11)	Test methods (12)	Remarks (13)
Uncabled optical fibre	5	IEC 60793-2-50		
Attenuation coefficient (cabled fibres)	5.2.1	According to DS	IEC 60793-1-40, method A, B or C	
at 1 310 nm at 1 550 nm and at 1 625 nm ¹ .		≤ 0,40 dB/km ≤ 0,35 dB/km ≤ 0,40 dB/km		
Attenuation discontinuities at 1 310 and 1 550 nm	5.2.2	≤ 0,10 dB	IEC 60793-1-40, method C	
Cabled fibre cut-off wavelength	5.3	$\lambda_{cc} < \lambda$ operational	IEC 60793-1-44, method B	
Fibre colouring	5.4	IEC 60304	Visual inspection	
Outer diameter including colouring	8.2.1.1	IEC 60793-2	IEC 60793-1-20, method D	

4.2.2 Single mode dispersion shifted (B2) optical fibre

Table 2 – Single mode dispersion shifted (B2) optical fibre

Characteristics (9)	IEC 60794-3 Clause/subclause (10)	Family requirements (11)	Test methods (12)	Remarks (13)
Uncabled optical fibre	5.1	IEC 60793-2		
Attenuation coefficient (cabled fibres)	5.2.1	According to DS	IEC 60793-1-40, method A, B or C	
at 1 550 nm	5.2.1	≤ 0,35 dB/km		
Attenuation discontinuities at 1 550 nm	5.2.2	≤ 0,10 dB/km	IEC 60793-1-40, method C	
Cabled fibre cut-off wavelength	5.3	$\lambda_{cc} < \lambda$ operational	IEC 60793-1-44, method B	
Fibre colouring	5.4	IEC 60304	Visual inspection	
Outer diameter including colouring	8.2.1.1	IEC 60793-2	IEC 60793-1-20, method D	

¹ Measurements at 1 625 nm are optional

4.2.3 Single mode non-zero dispersion (B4) optical fibre

Table 3 – Single mode non-zero dispersion (B4) optical fibre

Characteristics (9)	IEC 60794-3 Clause/subclause (10)	Family requirements (11)	Test methods (12)	Remarks (13)
Uncabled optical fibre	5.1	IEC 60793-2		
Attenuation coefficient (cabled fibres)	5.2.1	According to DS	IEC 60793-1-40, method A, B or C	
at 1 550 nm	5.2.1	$\leq 0,35$ dB/km		
at 1 625 ² nm	5.2.1	$\leq 0,40$ dB/km		
Attenuation discontinuities at 1 550 nm	5.2.2	$\leq 0,10$ dB/km	IEC 60793-1-40, method C	
Cabled fibre cut-off wavelength	5.3	$\lambda_{cc} < \lambda$ operational	IEC 60793-1-44, method B	
Fibre colouring	5.4	IEC 60304	Visual inspection	
Outer diameter including colouring	8.2.1.1	IEC 60793-2	IEC 60793-1-20, method D	

4.2.4 Single mode (B6) optical fibre

Table 4 – Single mode (B6) optical fibre

Characteristics (9)	IEC 60794-3 Clause/subclause (10)	Family requirements (11)	Test methods (12)	Remarks (13)
Uncabled optical fibre	5.1	IEC 60793-2		
Attenuation coefficient (cabled fibres)	5.2.1	According to DS	IEC 60793-1-40, method A, B or C	
at 1 550 nm	5.2.1	$\leq 0,30$ dB/km		
at 1 625 ³ nm	5.2.1	$\leq 0,40$ dB/km		
Attenuation discontinuities at 1 550 nm	5.2.2	$\leq 0,10$ dB/km	IEC 60793-1-40, method C	
Cabled fibre cut-off wavelength	5.3	$\lambda_{cc} < \lambda$ operational	IEC 60793-1-44, method B	
Fibre colouring	5.4	IEC 60304	Visual inspection	
Outer diameter including colouring	8.2.1.1	IEC 60793-2	IEC 60793-1-20, method D	

4.2.5 Multimode fibres

Under consideration.

² Measurements at 1 625 nm are optional

³ Measurements at 1 625 nm are optional

4.3 Sewer cable constructions

4.3.1 Cable for installation within conduits (previously fixed to the sewer wall)

**Table 5 – Characteristics – Cable for installation within conduits
(previously fixed to the sewer wall)**

Characteristics (9)	IEC 60794-3 Clause/subclause (10)	Family requirements (11)	Test methods (12)	Remarks (13)
Lay-up	7.2	According to DS	Visual inspection	
Sewer cable core	7.3	According to DS		
Filling compound (if used)		According to DS	Either IEC 60794-1-2, method E14 or IEC 60811-5-1, Clause 4 IEC 60811-5-1, Clause 5 IEC 60811-5-1, Clause 8	
Dry blocking compound	7.3	According to DS	Under consideration	
Strength member	7.4	According to DS	Visual inspection	
- central				
- peripheral				
Moisture barrier	7.5	According to DS		
Metallic tapes (if any):				
Outer cable sheath	7.6			
Material				
Minimum sheath thickness		According to DS	IEC 60811-1-1	
Outer cable diameter		According to DS	IEC 60811-1-1	
Optional protection		According to DS		
Sheath marking	7.7			
Configuration, dimensions		According to DS	Visual inspection	
Abrasion resistance		According to DS	IEC 60794-1-2, method E2B	Method 1 Steel needle diameter $d = 1,0$ mm load: 4 N Method 2 with: felt pad consisting of either 1) water soaked wool felt or 2) rayon felt with < 30 % wool Weight: > 450 g
Sheath abrasion	9.2.8	According to DS	IEC 60794-1-2, method E2A	
Sewer cable length			Under consideration	

4.3.2 Cable for direct installation into the sewer duct

Table 6 – Characteristics – Cable for direct installation into the sewer duct

Characteristics (9)	IEC60794-3 Clause/subclause (10)	Family requirements (11)	Test methods (12)	Remarks (13)
Lay-up	7.2	According to DS	Visual inspection	
Sewer cable core	7.3	According to DS		
Filling compound (if used)		According to DS	Either IEC 60794-1-2, method E14 or IEC 60811-5-1, Clause 4 IEC 60811-5-1, Clause 5 IEC 60811-5-1, Clause 8	
Dry blocking compound	7.3	According to DS	Under consideration	
Strength member	7.4	According to DS	Visual inspection	
- central				
- peripheral				
Moisture barrier	7.5	According to DS		
Metallic tapes:				
Outer cable sheath	7.6			
Material				
Minimum sheath thickness		According to DS	IEC 60811-1-1	
Outer cable diameter		According to DS	IEC 60811-1-1	
Optional protection		According to DS		
Sheath marking	7.7			
Configuration, dimensions		According to DS	Visual inspection	
Abrasion resistance		According to DS	IEC 60794-1-2 method E2B	Method 1 Steel needle diameter $d = 1,0 \text{ mm}$ load: 4 N Method 2 with: felt pad consisting of either 1) water soaked wool felt or 2) rayon felt with < 30 % wool Weight: > 450 g
Sheath abrasion	9.2.8	According to DS	IEC 60794-1-2 method E2A	
Sewer cable length			Under consideration	

4.3.3 Conduit construction

Table 7 – Characteristics – Conduit construction

Characteristics (9)	IEC 60794-3 as applicable Clause/subclause (10)	Family requirements (11)	Test methods (12)	Remarks (13)
Material(s)				
Conduit inner diameter		According to DS	IEC 60811-1-1	
Conduit outer diameter		According to DS	IEC 60811-1-1	
Corrugated conduit (metallic) - metallic tape thickness - corrugated depth - pitch Non-corrugated conduit Inner tube (if any): - tube wall thickness Moisture barrier (if any): - metallic tapes Additional outer sheath (if any): - thickness	Under consideration	Under consideration	Under consideration	Under consideration
Conduit length			Under consideration	

4.4 Installation and operating conditions

4.4.1 Tests applicable to cables/cable elements

Table 8 – Tests applicable to cables/cable elements

Characteristics (9)	IEC 60794-3 Clause/subclause (10)	Family requirements (11)	Test methods (12)	Remarks (13)
General requirements Tests applicable to loose tubes:	8.1	Agreement between customer and supplier		
Bend test	8.2.1.2	According to DS	IEC 60794-1-2, method G1	
Tube kinking	8.2.2.1	According to DS	IEC 60794-1-2, method G7	
Tests applicable to ribbons:				
- dimensions	8.2.3.1	IEC 60794-3, Table 1	IEC 60794-3, 8.2.3.1	
- separability of individual fibres from ribbon	8.2.3.2.1	IEC 60794-3, 7.2.3.2.1 or according to DS	IEC 60794-1-2, method G5 or according to DS	
- ribbon stripping	8.2.3.2.2	According to DS		
- torsion	8.2.3.2.3	According to DS	IEC 60794-1-2, method G6	

4.4.2 Installation conditions

Under consideration.

4.5 Mechanical and environmental tests

4.5.1 Conduits

4.5.1.1 Tests applicable

Tests listed in the following table are those relevant to IEC 60794-3 as applicable for conduits.

Table 9 – Conduits – Tests applicable

Characteristics (9)	IEC 60794-3 Clause/subclause (10)	Family requirements (11)	Test methods (12)	Remarks (13)
Tensile performance	9.1	4.5.1.2.2	IEC 60794-1-2, methods E1A and E1B	See 5.6.2.1
Installation capability (selection from the following)	9.2			
- bending under tension	9.2.1	According to DS	IEC 60794-1-2, method E18	
- repeated bending	9.2.2	na	IEC 60794-1-2, method E6	
- impact	9.2.3	4.5.1.2.5	IEC 60794-1-2, method E4	
- kink	9.2.4	4.5.1.2.3	IEC 60794-1-2, method E10	
- torsion	9.2.5		IEC 60794-1-2, method E7	
Conduit bend	9.3	According to DS	IEC 60794-1-2, method E11	See 5.6.2.5
Crush	9.4	4.5.1.2.4	IEC 60794-1-2, method E3	See 5.6.2.6
Flexibility		4.5.1.2.6		
Ageing	9.6	na		
- finished conduit	9.6.2	Under consideration		
Pressure		4.5.1.2.1		
Induced voltage (for tubes with metallic elements)	9.9	Under consideration		
Chemical exposure (fluid and contaminant)	Under consideration	4.5.1.2.7	Under consideration	

4.5.1.2 Details of family requirements and test conditions for conduits

Tests shall be selected from those of Table 9 and the following described hereinafter:

4.5.1.2.1 Pressure

a) Family requirements

Under visual examination, without magnification, there shall be no damage to the conduit.

b) Test conditions

Method: under consideration

All conduits shall resist an air pressure of at least $[2,5 \times \text{the installation pressure}]$ at a temperature of 20 °C for a period of 0,5 h.

Additionally, the same performance shall be demonstrated after a sample of conduit has been maintained at 60 °C for a period of 12 weeks.

All conduits shall resist a proof test pressure of at least $[1,3 \times \text{the installation pressure}]$ at a temperature of 20 °C for a period of 24 h, after tensile and bending tests.

4.5.1.2.2 Tensile performance

a) Family requirements

Under visual examination without magnification, there shall be no damage and the diameter shall not change by more than $\times \%$.

b) Test conditions

Method: under consideration

Tube length under tension: under consideration

Tensile load on tube: under consideration

Diameter of test pulleys: under consideration

4.5.1.2.3 Kink

a) Family requirements

Under visual examination, without magnification, there shall be no kink to the conduits.

b) Test conditions

Method: IEC 60794-1-2, E10

Minimum diameter: 20 times the outer diameter of the conduit

4.5.1.2.4 Crush

a) Family requirements

Under visual examination, without magnification, there shall be no damage to the conduit. There shall be no residual deformation greater than 15 % of the tube diameter, no splitting or permanent damage after removing the load. The imprint of the anvil on the jacket is not considered as mechanical damage.

b) Test conditions

Method: IEC 60794-1-2, E3

Sample length: 250 mm

load (plate/plate): $50 \times d_c$ (N) (d_c in mm) or 450 N whichever is lower

Duration time: 60 s

Recovery time: 1 h

4.5.1.2.5 Impact

a) Family requirements

Under visual examination without magnification, there shall be no damage to the conduit. There shall be no residual deformation greater than 15 % of the conduit diameter, no splitting or permanent damage. The imprint of the striking surface on the tube is not considered mechanical damage.

b) Test conditions

Method: under consideration
Striking surface radius: 10 mm
Impact energy: 1 J
Recovery time: 1 h
Number of impacts: one in 3 different places spaced not less than 500 mm apart

4.5.1.2.6 Flexibility

a) Family requirements

The outer and inner diameter of the conduits shall show, under visual examination without magnification, no damage and no reduction of diameter greater than 15 %.

b) Tests conditions

Method: ten turns of the conduit shall be wrapped tightly and secured around a mandrel of diameter 12 times the outer diameter of the conduit itself

Duration: 30 min

4.5.1.2.7 Chemical exposure

a) Family requirements

Under consideration

b) Test conditions

Under consideration

4.5.2 Cable for installation within conduits (previously fixed to the sewer wall)**4.5.2.1 Tests applicable****Table 10 – Optical fibre cable – Tests applicable**

Characteristics (9)	IEC 60794-3 Clause/subclause (10)	Family requirements (11)	Test methods (12)	Remarks (13)
Tensile performance	9.1	4.5.2.2.1	IEC 60794-1-2, methods E1A and E1B	
Installation capability (selection from the following)	9.2			
- bending under tension	9.2.1	According to DS	IEC 60794-1-2, method E18	
- repeated bending	9.2.2	4.5.2.2.2	IEC 60794-1-2, method E6	
- impact	9.2.3		IEC 60794-1-2, method E4	
- kink	9.2.4	According to DS	IEC 60794-1-2, method E10	
- torsion - blowing	9.2.5	4.5.2.2.3 According to DS	IEC 60794-1-2, method E7	
Cable bend	9.3	4.6.2.2.4	IEC 60794-1-2, method E11	
Crush	9.4	4.5.2.2.5	IEC 60794-1-2, method E3	
Impact	9.2.3	4.5.2.2.6	IEC 60794-1-2, method E4	
Temperature cycling	9.5	4.5.2.2.7	IEC 60794-1-2, method F1	
Ageing	9.6			
- coating adhesion stability	9.6.1	According to DS	IEC 60794-1-2, method E5	
- finished cable	9.6.2	Under consideration		
Water penetration	9.7	According to DS	IEC 60794-1-2, method F5B	
Pneumatic resistance (for unfilled cable)	9.8	According to DS (under consideration)	IEC 60794-1-2, method F6	
Induced voltage (for cables with metallic elements)	9.9	Under consideration		
Chemical exposure (fluid and contaminant)	Under consideration	Under consideration	Under consideration	

4.5.2.2 Details of family requirements and test conditions for sewer cable tests

The expression of "no change in attenuation" means that any change in measurement value, either positive or negative, within the uncertainty of measurement shall be ignored. The uncertainty of measurement for this standard shall be < 0,05 dB for attenuation.

Tests shall be selected from those given in Table 10 and the following described hereinafter:

4.5.2.2.1 Tensile performance⁴

a) Family requirements

Under long term tensile load (T_L) the fibre strain shall not exceed 20 % of the fibre proof strain and there shall be no change in attenuation during the test. Under installation load (T_M) the fibre strain shall not exceed 60 % of the fibre proof strain and the attenuation change during test shall be measured and recorded. Other criteria may be agreed between the customer and the supplier.

Under visual examination without magnification, there shall be no damage to the sheath or to the cable elements.

There shall be no change after test in attenuation when measured in the 1 550 nm region or at the operational wavelength when specified by the customer, at room temperature.

b) Test conditions

Method:	IEC 60794-1-2, E1A and E1B
Cable length under tension:	not less than 50 m. Taking into account the measurement accuracy and end effects, shorter lengths may be used by agreement between the customer and the supplier
Fibre length:	finished cable length
Tensile load on cable:	long term tensile load (T_L) and installation load (T_M) Other loads may be applied in accordance with particular user conditions
Diameter of test pulleys:	1 m but not less than the minimum loaded bending diameter specified for the cable
T_M :	equivalent to weight of 1 km of sewer cable or 50 N whichever is greater
T_L :	equivalent to weight of 500 m of sewer cable or 25 N whichever is greater (ffs)

4.5.2.2.2 Repeated bending

a) Family requirements

Under visual examination without magnification, there shall be no damage to the sheath and to the cable elements.

b) Test conditions

Method:	IEC 60794-1-2, E6
Bending radius:	20 d or 30 mm whichever is greater
Load:	adequate to assure uniform contact with the mandrel
Number of cycles:	25
Duration of cycle:	approximately 2 s

4.5.2.2.3 Torsion

a) Family requirements

⁴ Cables for blowing in ducts with thin sheath constructions are not designed for long distance pulling (i.e. 300 m)

Under visual examination without magnification, there shall be no damage to the sheath or to the cable elements.

The variation on attenuation for each fibre shall be less than or equal to 0,10 dB at 1 550 nm, or at the operational wavelength when specified by the customer.

There shall be no permanent change in attenuation after the test.

b) Test conditions

Method: IEC 60794-1-2, E7

Length under test: 2 m

Number of turns: one half turn (through 180°) over the length of 2 m in each direction

Number of cycles: 5

4.5.2.2.4 Bend

a) Family requirements

There shall be no change in attenuation when measured in the 1 550 nm region or at the operational wavelength when specified by the customer, at room temperature.

The change in attenuation when tested at –30 °C shall be ≤ 0,1 dB if required.

b) Test conditions

Method: IEC 60794-1-2, E11

Diameter of mandrel: ≤ 40 *d* or 60 mm whichever is greater

Number of turns/helix: 4

Number of cycles: 3

4.5.2.2.5 Crush

a) Family requirements

Immediately after removal of the load, there shall be no increase in attenuation when measured in the 1 550 nm region or at the operational wavelength when specified by the customer.

Under visual examination, there shall be no damage to the sheath or to the cable elements after removing the load. The imprint of the plate or mandrel on the sheath is not considered mechanical damage.

b) Test conditions

Method: IEC 60794-1-2, E3

Load (plate/plate): 450 N

Duration of load: 1 min

4.5.2.2.6 Impact

a) Family requirements

The cable shall be tested in accordance with IEC 60794-1-2, method E4.

b) Test conditions:

Striking surface radius: 10 mm

Impact energy: 1J with striking surface radius of 10 mm

Number of impacts: one in 3 different places spaced not less than 500 mm apart

Under visual examination without magnification, there shall be no damage to the sheath or to the cable elements. The imprint of the anvil on the sheath is not considered as mechanical damage.

There shall be no change after test in attenuation when measured in the 1 550 nm region or at the operational wavelength when specified by the customer, at room temperature.

4.5.2.2.7 Temperature cycling

a) Family requirements

During the last cycle, there shall be no change in attenuation between the initial room temperature measurement and T_{A1} or T_{B1} . For 1 625 nm applications, performance criteria shall be mutually agreed upon between the customer and supplier.

During the last cycle, the attenuation change from the room temperature measurement and T_{A2} or T_{B2} shall be $< 0,15$ dB/km at 1 550 nm. For 1 625 nm applications, performance criteria shall be mutually agreed upon between the customer and supplier. On completion of the test, there shall be no change in attenuation.

Definition of a cycle: T_{B2} to T_{A2}

Definition of the last cycle: T_{A2} , T_{A1} , T_{B1} , T_{B2} with a final measurement at room temperature.

b) Test conditions

Sample length: finished cable length of at least 1 000 m

High temperature, T_{B2} : +60 °C to +70 °C, depending on customer requirements

High temperature, T_{B1} : +30 °C to +60 °C, depending on customer requirements

Low temperature, T_{A1} : –10 °C to –15 °C, depending on customer requirements

Low temperature, T_{A2} : T_{A1} to –40 or –45 °C, depending on customer requirements

Rate of heating: sufficiently slow that the effect of changing the cooling temperature does not cause temperature shock

t_1 : temperature cycling test dwell time until a stable temperature is reached

Number of cycles: 2, but additional cycles may be required in accordance with particular customer requirements

4.5.2.2.8 Chemical exposure

a) Family requirements

Under consideration

b) Test conditions

Under consideration

4.5.3 Cables for direct installation into the sewer duct

4.5.3.1 Tests applicable

Table 11 – Tests applicable

Characteristics (9)	IEC 60794-3 Clause/subclause (10)	Family requirements (11)	Test methods (12)	Remarks (13)
Tensile performance	9.1	4.5.3.2.1 and according to DS	IEC 60794-1-2, methods E1A and E1B	
Installation capability (selection from the following)	9.2			
- bending under tension	9.2.1	According to DS	IEC 60794-1-2, method E18	
- repeated bending	9.2.2	4.5.3.2.2	IEC 60794-1-2, method E6	
- impact	9.2.3	4 5.3.2.6	IEC 60794-1-2, method E4	
- kink	9.2.4	According to DS	IEC 60794-1-2, method E10	
- torsion	9.2.5	4.5.3.2.3	IEC 60794-1-2, method E7	
Cable bend	9.3	According to DS and 4.5.3.2.4	IEC 60794-1-2, method E11	
Crush	9.4	According to DS and 4.5.3.2.5	IEC 60794-1-2, method E3	
Temperature cycling	9.5	According to DS and 4.5.3.2.7	IEC 60794-1-2, method F1	
Ageing	9.6			
- coating adhesion stability	9.6.1	According to DS	IEC 60794-1-2, method E5	
- finished cable	9.6.2	Under consideration		
Water penetration	9.7	According to DS	IEC 60794-1-2, method F5A, F5B	
Pneumatic resistance (for unfilled cable)	9.8	According to DS (under consideration)	IEC 60794-1-2, method F6	
Induced voltage (for cables with metallic elements)	9.9	Under consideration		
Mould growth	Under consideration	Under consideration	Under consideration	
Chemical exposure (fluid and contaminant)	Under consideration	Under consideration	Under consideration	

4.5.3.2 Details of family requirements and test conditions for sewer cables

The expression of "no change in attenuation" means that any change in measurement value, either positive or negative, within the uncertainty of measurement shall be ignored. The uncertainty of measurement for this standard shall be < 0,05 dB for attenuation.

Tests shall be selected from those given in Table 11 and the following described hereinafter:

4.5.3.2.1 Tensile performance⁵

a) Family requirements

Under long term tensile load (T_L) the fibre strain shall not exceed 20 % of the fibre proof strain and there shall be no change in attenuation during the test. Under installation load (T_M) the fibre strain shall not exceed 60 % of the fibre proof strain and the attenuation change during test shall be measured and recorded. Other criteria may be agreed between the customer and the supplier.

Under visual examination without magnification, there shall be no damage to the sheath or to the cable elements.

There shall be no change after test in attenuation when measured in the 1 550 nm region or at the operational wavelength when specified by the customer, at room temperature.

b) Test conditions

Method:	IEC 60794-1-2, E1A and E1B
Cable length under tension:	not less than 50 m. Taking into account the measurement accuracy and end effects, shorter lengths may be used by agreement between the customer and the supplier
Fibre length:	finished cable length
Tensile load on cable:	long term tensile load (T_L) and installation load (T_M) Other loads may be applied in accordance with particular user conditions
Diameter of test pulleys:	1 m but not less than the minimum loaded bending diameter specified for the cable
T_M :	equivalent to weight of 1 km of sewer cable or 50 N whichever is greater
T_L :	equivalent to weight of 500 m of sewer cable or 25 N whichever is greater (ffs)

4.5.3.2.2 Repeated bending

a) Family requirements

Under visual examination without magnification, there shall be no damage to the sheath and to the cable elements.

b) Test conditions

Method:	IEC 60794-1-2, E6
Bending radius:	20 d or 30 mm whichever is greater
Load:	adequate to assure uniform contact with the mandrel
Number of cycles:	25
Duration of cycle:	approximately 2 s

4.5.3.2.3 Torsion

a) Family requirements

Under visual examination without magnification, there shall be no damage to the sheath or to the cable elements.

The variation on attenuation for each fibre shall be less than, or equal to, 0,10 dB at 1 550 nm, or at the operational wavelength when specified by the customer.

There shall be no permanent change in attenuation after the test.

b) Test conditions

Method:	IEC 60794-1-2, E7
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⁵ Cables for blowing in ducts with thin sheath constructions are not designed for long distance pulling (i.e. 300 m)

Length under test: 2 m

Number of turns: one half turn (through 180°) over the length of 2 m in each direction

Number of cycles: 5

4.5.3.2.4 Bend

a) Family requirements

There shall be no change in attenuation when measured in the 1 550 nm region or at the operational wavelength when specified by the customer, at room temperature.

The change in attenuation when tested at –30 °C shall be $\leq 0,1$ dB.

b) Test conditions

Method: IEC 60794-1-2-E11

Diameter of mandrel: $\leq 40 d$ or 60 mm whichever is greater

Number of turns/helix: 4

Number of cycles: 3

4.5.3.2.5 Crush

a) Family requirements

Immediately after removal of the load, there shall be no increase in attenuation when measured in the 1 550 nm region or at the operational wavelength when specified by the customer.

Under visual examination, there shall be no damage to the sheath or to the cable elements after removing the load. The imprint of the plate or mandrel on the sheath is not considered mechanical damage.

b) Test conditions

Method: IEC 60794-1-2, E3

Load (plate/plate): 2 500 N

Duration of load: 1 min

4.5.3.2.6 Impact

a) Family requirements

The cable shall be tested in accordance with IEC 60794-1-2, method E4.

b) Test conditions

Striking surface radius: 10 mm

Impact energy: 10 J with striking surface radius of 10 mm

Number of impacts: one in 3 different places spaced not less than 500 mm apart

Under visual examination without magnification, there shall be no damage to the sheath or to the cable elements. The imprint of the anvil on the sheath is not considered as mechanical damage.

There shall be no change after test in attenuation when measured in the 1 550 nm region or at the operational wavelength when specified by the customer, at room temperature.

4.5.3.2.7 Temperature cycling

a) Family requirements

During the last cycle, there shall be no change in attenuation between the initial room temperature measurement and T_{A1} or T_{B1} . For 1 625 nm applications, performance criteria shall be mutually agreed upon between the customer and supplier.

During the last cycle, the attenuation change from the room temperature measurement and T_{A2} or T_{B2} shall be $< 0,15$ dB/km at 1 550 nm. For 1 625 nm applications,

performance criteria shall be mutually agreed upon between the customer and supplier. On completion of the test, there shall be no change in attenuation.

Definition of a cycle: T_{B2} to T_{A2}

Definition of the last cycle: T_{A2} , T_{A1} , T_{B1} , T_{B2} with a final measurement at room temperature.

b) Test conditions

Sample length: finished cable length of at least 1 000 m

High temperature, T_{B2} : +60 °C to +70 °C, depending on customer requirements

High temperature, T_{B1} : +30 °C to +60 °C, depending on customer requirements

Low temperature, T_{A1} : –10 °C to –15 °C, depending on customer requirements

Low temperature, T_{A2} : T_{A1} to –40 or –45 °C, depending on customer requirements

Rate of heating: sufficiently slow that the effect of changing the cooling temperature does not cause temperature shock

t_1 : temperature cycling test dwell time until a stable temperature is reached

Number of cycles: 2, but additional cycles may be required in accordance with particular customer requirements

4.5.3.2.8 Chemical exposure

a) Family requirements

Fluids and gas for which the cable shall be tested will be agreed between the customer and the supplier, preferably taken from IEC 62362/TR

At the conclusion of the recovery period, the cables shall meet the requirements of the relevant detail or sectional specification for the following properties, unless otherwise specified:

- visual inspection of all construction elements;
- tensile strength and elongation at break of sheath;
- no change in attenuation.

b) Test method

IEC 62362/TR, 4.3.4 and 4.3.5

Annex A (informative)

Blank detail specification

A.1 Sewer cables description

A.1.1 Cable for installation within conduits (previously fixed to the sewer wall)

Such a sewer cable has to be blown or pulled into the conduits described in 4.1.2.

Table A.1 – Sewer optical fibre cable description – Within conduits

(1) Prepared by		(2) Document No.: Issue: Date:
(3) Available from	(4) Generic specifications: Sectional specification:	IEC 60794-1-1 and IEC 60794-1-2 IEC 60794-3
(5) Additional references:		
Construction - tube - unfilled - tube - filled Inner sheath Outer sheath Additional armouring - non-metallic armouring - metallic armouring Additional outer sheath Marking identification - customer requirement - identification of supplier		Additional remarks
(8) Application information:		
Maximum outer diameter (d)		mm
Rated maximum tensile load		N
Minimum bending radius for no-load bending		mm or $n \times d$
Minimum bending radius for rated-load bending		mm or $n \times d$
Temperature range:		
- transport and storage		°C
- installation		°C
- operation		°C
Delivery length		
- typical		m
- nominal/tolerances		-0 + 1%

A.1.2 Cables for direct installation into the sewer duct

Such cables are directly screwed on the sewer wall by means of proper “hooks” using a special designed robot, or spanned – similar to aerial cables – by means of anchors between two man-holes limiting sag and interference with lateral tubes, or laid and fixed on the ground of the sewer mainly by gravity and a certain number of anchor devices.

Table A.2 – Sewer optical fibre description – Direct installation

(1) Prepared by		(2) Document No.: Issue: Date:
(3) Available from	(4) Generic specifications: Sectional specification:	IEC 60794-1-1 and IEC 60794-1-2 IEC 60794-3
(5) Additional references:		
Construction - tube - unfilled - tube - filled Inner sheath Outer sheath Additional armouring - non-metallic armouring - metallic armouring Additional outer sheath Marking identification - customer requirement - identification of supplier		Additional remarks
(8) Application information:		
Maximum outer diameter (<i>d</i>)		mm
Rated maximum tensile load		N
Minimum bending radius for no-load bending		mm or $n \times d$
Minimum bending radius for rated-load bending		mm or $n \times d$
Temperature range:		
- transport and storage		°C
- installation		°C
- operation		°C
Delivery length		
- typical		m
- nominal/tolerances		-0 + 1%

A.1.3 Conduit description

Such conduit is directly attached on the sewer wall by means of proper fittings. In the case of use of stainless-steel-conduits, the conduit is fixed by clamps to spring loaded stainless-steel rings installed into the cleaned and inspected sewer tube, distanced about 1,5 m, using a special installation robot.

Table A.3 – Conduit description

(1)	Prepared by		(2) Document No.: Issue: Date:
(3)	Available from	(4)	Generic specifications: IEC 60794-1-1 and IEC 60794-1-2 Sectional specification: IEC 60794-3 (all as applicable to conduits)
(5)	Additional references:		
Construction			
- corrugated conduit (metallic)			
- non-corrugated conduit			
Additional armouring			
- metallic			
- non-metallic			
Additional outer sheath			
Marking identification			
- customer requirement			
- identification of the supplier			
(8)	Application information:		
Maximum outer diameter (<i>d</i>)		mm	
Rated maximum tensile load		N	
Minimum bending radius for no-load bending		mm	
Minimum bending radius for rated-load bending		mm	
Temperature range:		°C	
- transport and storage		°C	
- installation		°C	
- operation			
Delivery tube length			
- typical		m	
- nominal/tolerances		-0 + 1 %	

Annex B (informative)

OF cables for non-man accessible sewers

Table B.1 – Characteristics for optical fibre cables within non-man accessible sewers

Cable characteristics versus cable installation methods	Screwed on the sewer wall	Installed within conduits previously fixed to the sewer wall	Spanning between manholes (type “aerial” cables)	Laying on the ground of the sewer
International specification No.	IEC 60794-3-10	IEC 60794-3-10	IEC 60794-3-20 as applicable IEC 60794-4 as applicable	IEC 60794-3-20 as applicable IEC 60794-4 as applicable
Dimensions maximum outer diameter ^a	< yy mm	< yy mm (conduit)	< yy mm	< yy mm
Additional requirements besides the International specification				
Preferential environmental protection	Metallic corrugated steel / SPL / APL / Al sheath	Metallic (conduit) corrugated (stainless) steel / SPL sheath	Metallic (stainless) steel / tube(s) / SPL / APL / Al sheath	Metallic corrugated steel / SPL / APL / Al sheath
Need for chemicals resistance	YES	YES	YES	YES
Rodent protection	YES	No	YES	YES
^a Cable/conduit maximum outer diameter should be agreed case by case between the cable supplier and the customer (sewer owner, etc.).				

Annex C (informative)

Examples of conduits and sewer cables

C.1 Cables for installation within conduits

C.1.1 Dielectric sewer cables

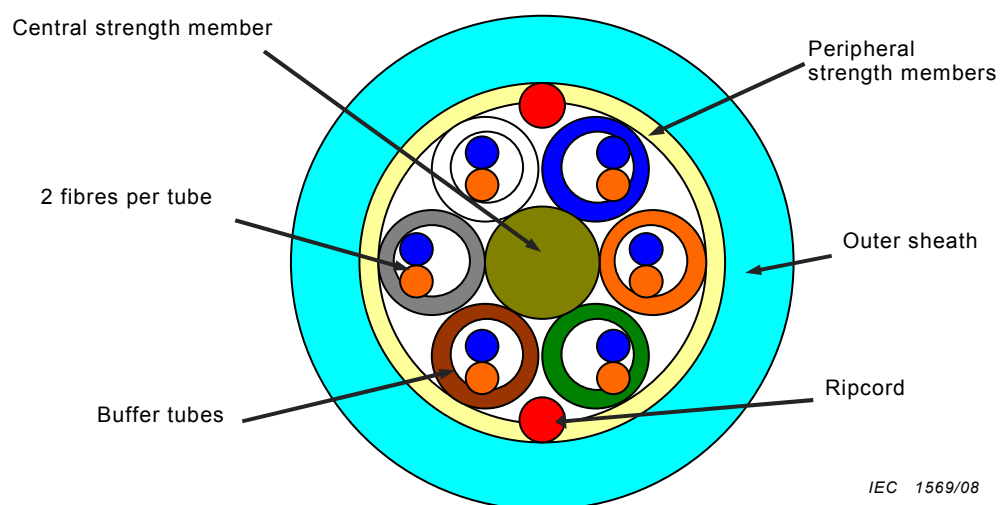


Figure C.1 – Dielectric optical fibre sewer cable

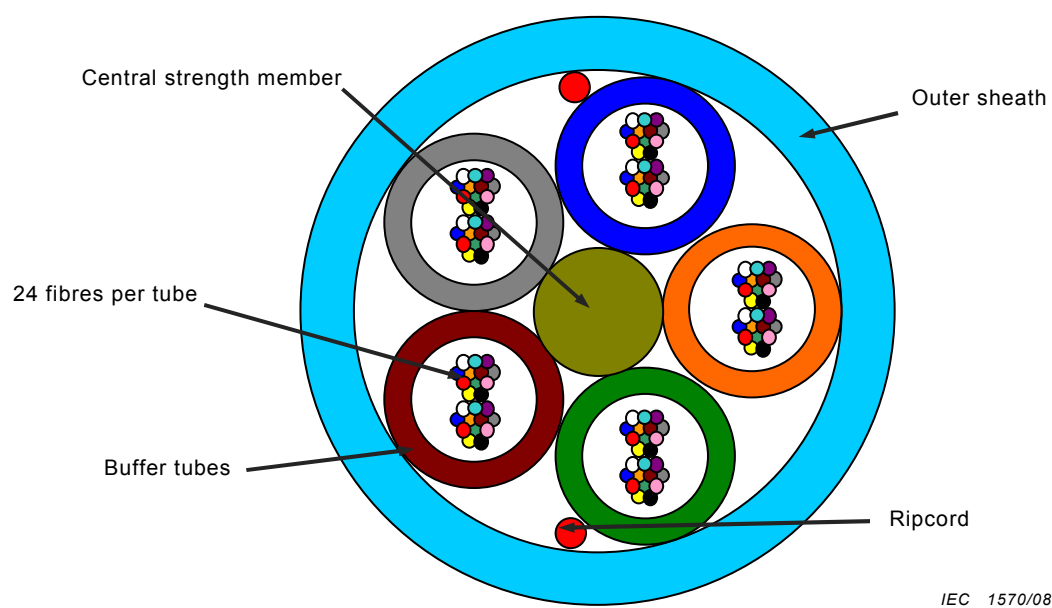
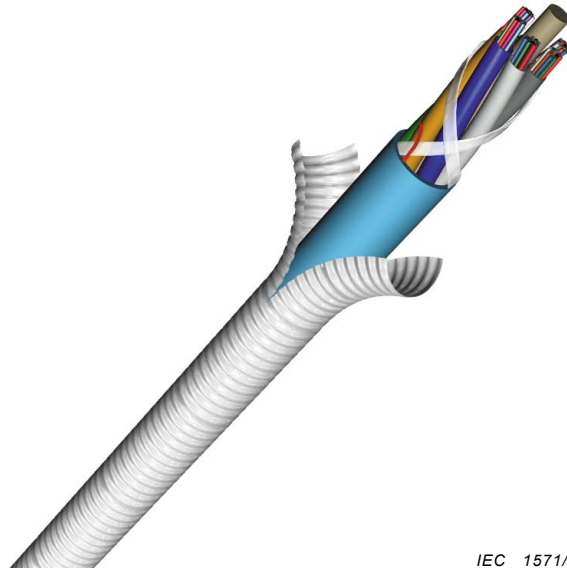


Figure C.2 – Dielectric optical fibre sewer cable

C.1.2 Sewer cable installed within a conduit



IEC 1571/08

Figure C.3 – Optical fibre sewer cable within a conduit

C.2 Cables for direct installation into the sewer duct

C.2.1 Cables to be screwed to the sewer inner wall

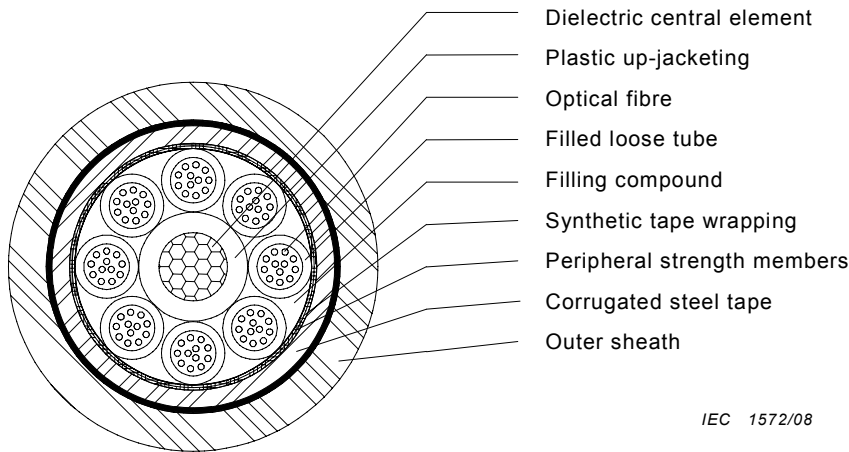


Figure C.4 – Optical fibre sewer cable for direct installation – peripheral strength members

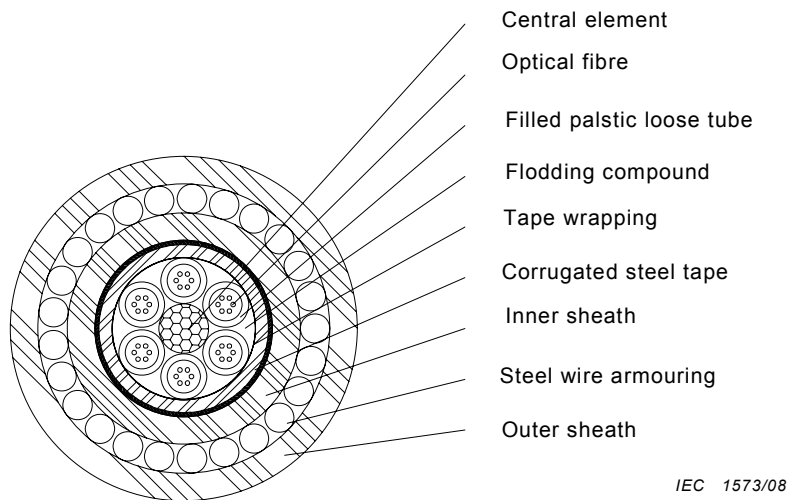


Figure C.5 – Optical fibre sewer cable for direct installation – steel wire armoring

C.2.2 Cables for spanning between manholes, similarly to aerial cables

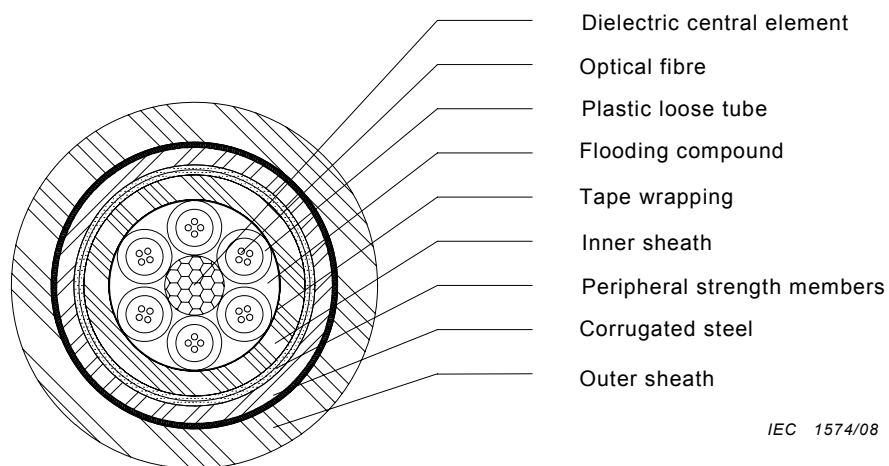


Figure C.6 – Optical fibre sewer cable for spanning – peripheral strength members

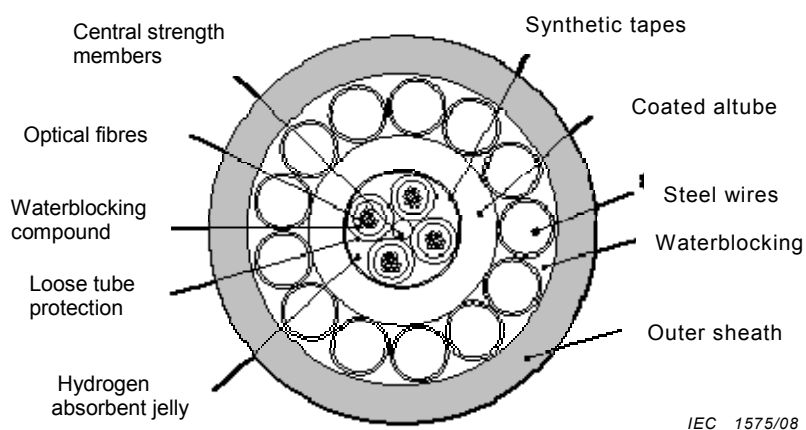


Figure C.7 – Optical fibre sewer cable for spanning – steel wire armouring

C.2.3 Cables for laying on the ground of the sewer

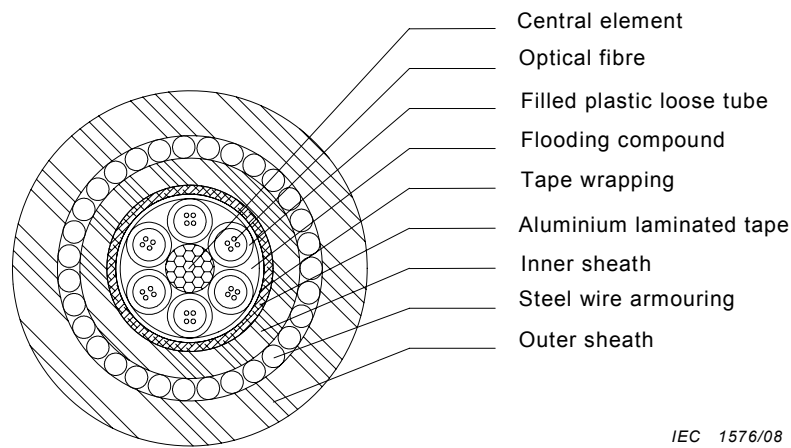


Figure C.8 – Optical fibre sewer cable for laying – aluminium tape

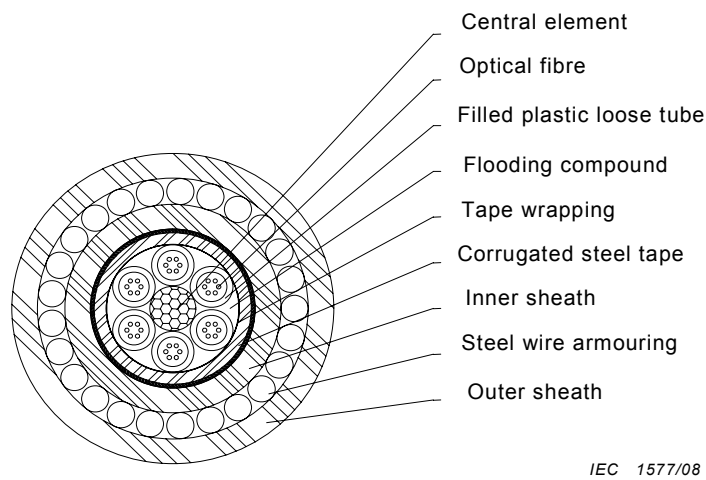
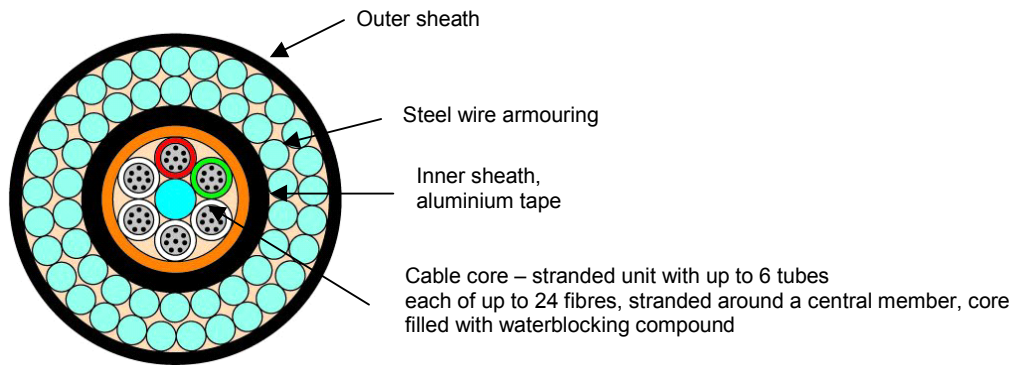


Figure C.9 – Optical fibre sewer cable for laying – corrugated steel



Sample of 144 OF

IEC 1578/08

Figure C.10 – Optical fibre sewer cable for laying – 2-layer-steel wire armouring

Annex D (informative)

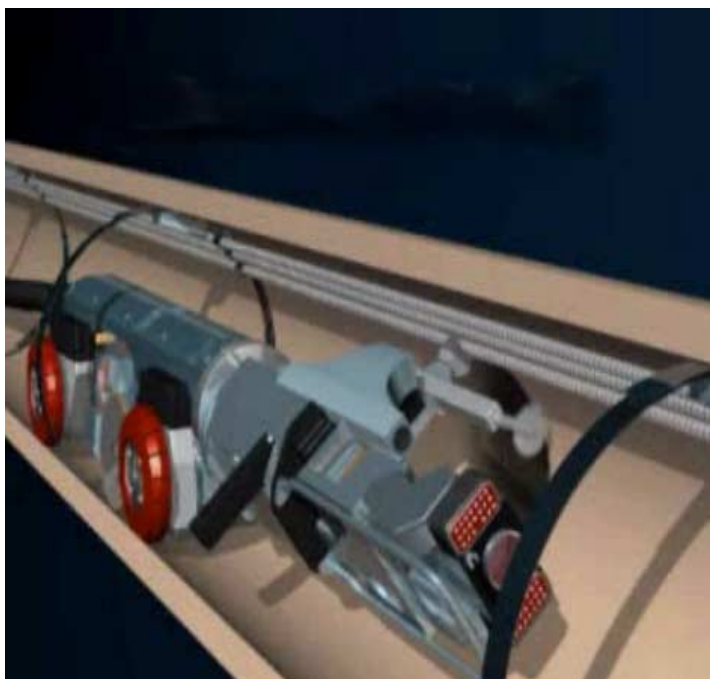
Examples of installation schemes

D.1 Installation within conduits

Conduits are fixed, for example by clamps to spring loaded stainless-steel rings installed into the sewer tube using a special designed robot.

Sewer cables will be blown or pulled later within such conduits.

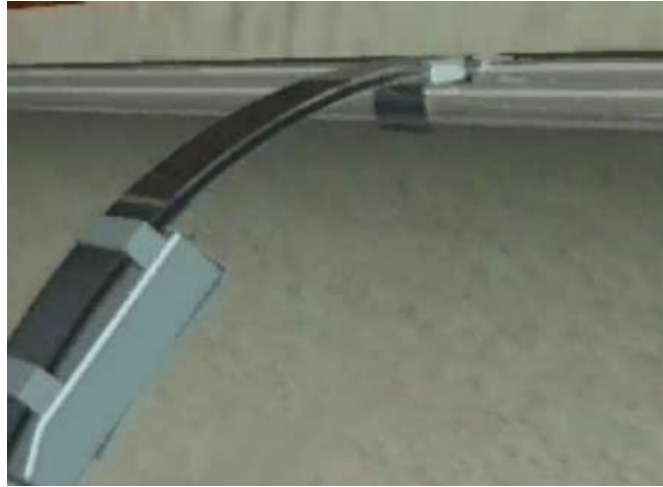
D.1.1 Conduit robotized installation



IEC 1579/08

Figure D.1 – Conduit robotized installation

D.1.2 Spring loaded stainless-steel ring

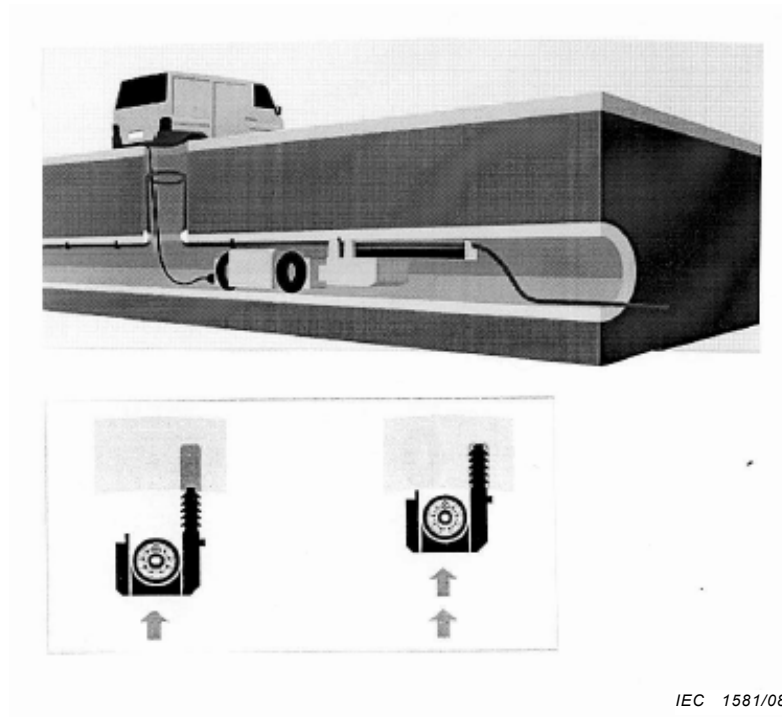


IEC 1580/08

Figure D.2 – Spring loaded stainless-steel ring – conduit fastening

D.2 Installation directly into the sewer

D.2.1 Cable screwed to the inner sewer wall by means of an installation robot



IEC 1581/08

Figure D.3 – Schematic drawing robotized installation – Drilling

D.2.2 Cable spanned between manholes, similarly to aerial cables

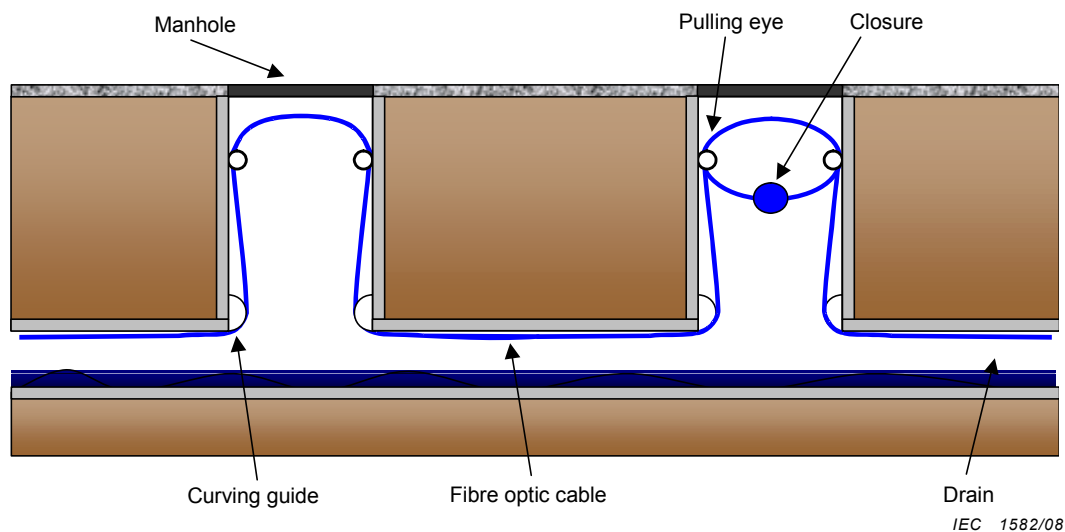


Figure D.4 – Schematic drawing – Spanning of optical fibre cables within sewers

D.2.3 Cable laid on the ground of the sewer

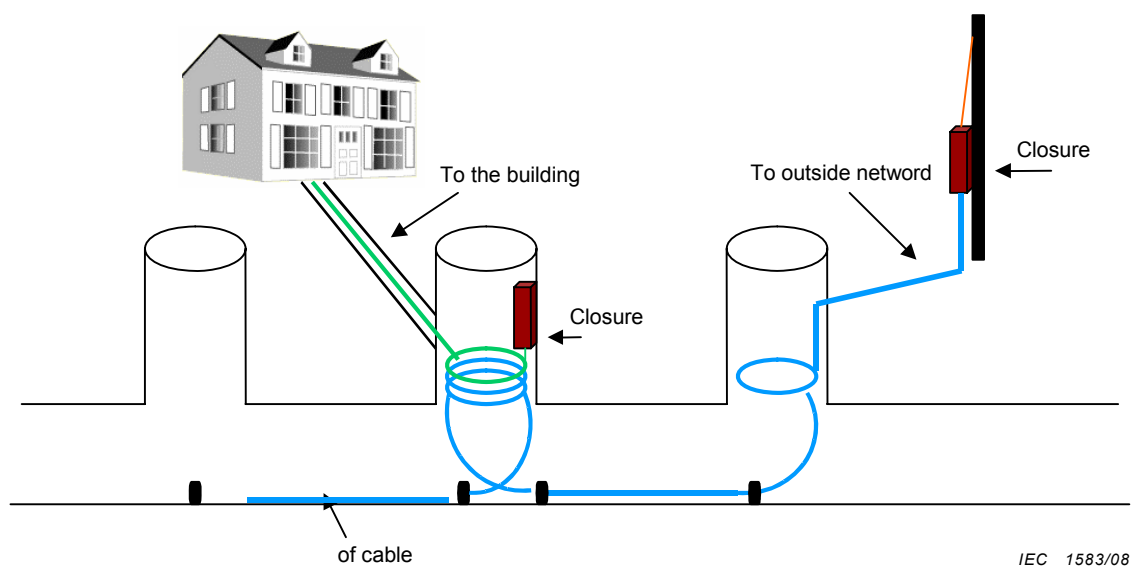


Figure D.5 – Schematic drawing – laying on the ground of optical fibre cables within sewers

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